

CE 404: Analysis of Structures

Course Code	CE 404
Course Name	Analysis of Structures
L-T-P-C	3-0-0-3
Prerequisites	CE 301: Strength of Materials and Structures/Equivalent
Intended for	UG Civil Engineering
Mutual Exclusion	None
Approval	50th BoA (Jul 2023), in the B.Tech Civil Engineering curriculum; 52nd BoA (Nov 2023)

Course Contents

- **Introduction:** Statically determinate vs statically indeterminate structures, Degrees of static and kinematic indeterminacy, Review on the methods of analysis of determinate structures. (3 Lectures)
- **Energy methods of structural analysis:** Basic concept, Energy relations with structural theory, Virtual work and its application, Energy principles based on displacement and force fields. (5 Lectures)
- **Analysis using force method:** Analysis of indeterminate beams and frames, Law of reciprocal deflections, Theorem of least work, Reactions due to yielding of supports, Analysis of indeterminate trusses using force method. (6 Lectures)
- **Analysis using the displacement method:** Slope-deflection method: Analysis of continuous beams, analysis of rigid frames with and without side sway; Moment distribution method: Analysis of indeterminate beams subjected to loads and uneven settlement of supports, analysis of rigid frames with and without side sway. (8 Lectures)
- **Three-moment theorem:** Derivation and application for analysis of continuous beams subjected to loads and supports settlement. (4 Lectures)
- **Arches:** Introduction, Basic mechanics, Three hinged arch, Two hinged arch, Tied arch and bowstring girder, Analysis of cables and suspension bridges. (5 Lectures)
- **Approximate analysis of statically indeterminate structures:** Analysis of trusses, Gravity loaded building frames, Portal and cantilever methods for laterally loaded building frames. (4 Lectures)
- **Moving load and Influence lines:** Introduction to the methodology and application to indeterminate structures. (2 Lectures)
- **Advanced concepts:** Brief introduction to matrix stiffness method of structural analysis and exposure to relevant software's, Basic introduction to plastic analysis: plastic hinge, shape factor, collapse mechanism, Methods of analysis: static and kinematic. (5 Lectures)

Text Books

1. Hibbeler, RC, **Structural Analysis**, 9th edition, Pearson Education, 2017.
2. Reddy, C.S., **Basic Structural Analysis**, Tata McGraw Hill, 2001.

References

1. Wang, C.K., **Intermediate Structural Analysis**, Tata McGraw Hill, New Delhi, 2010.
2. Ghali, A., and Neville, A.M., **Structural Analysis: A Unified Classical and Matrix Approach**, 7th Edition, CRC Press, 2017.
3. Negi, LS, and Jangid, R.S., **Structural Analysis**, Tata McGraw Hill, 1997.
4. Gupta, S. P., Gupta, R, and Pandit, GS, **Theory of Structures**, Tata McGraw-Hill, 1999.
5. Menon, D, **Advanced Structural Analysis**, Narosa Publishing House, 2015.
6. Kassimali, A., **Structural Analysis**, 6th Edition, Cengage Learning, 2018.